

Resistivity

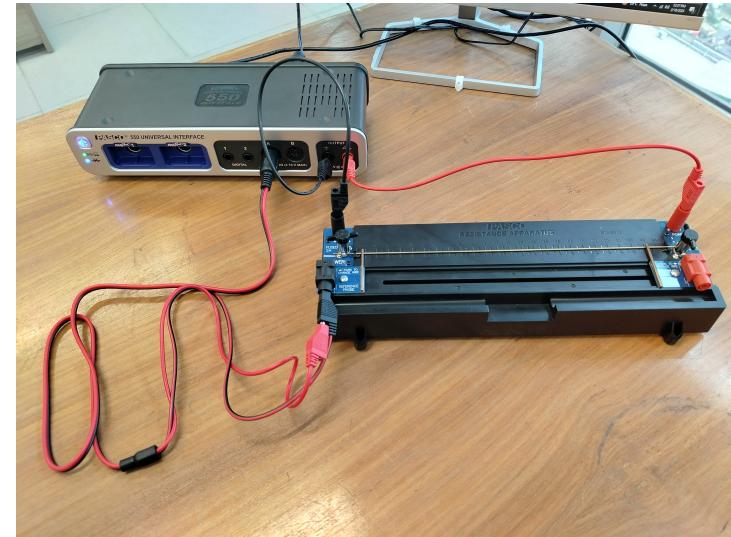
Equipment

Included:

1	Resistance Apparatus	EM-8812
1	Voltage Sensor	UI-5100
1	Patch cords (set of 5)	SE-9750
NOT INCLUDED, BUT REQUIRED:		
1	550 Universal Interface	UI-5001
1	Micrometer	SE-7337

Introduction

The resistivity of different metals is determined by finding the resistance of wires of a known diameter as a function of their length. It is also shown that the resistance of a wire of fixed length is inversely proportional to its cross-sectional area.



Theory

If a current (I) is flowing through a wire, the voltage drop (V) across a certain length of wire with resistance R is given by Ohm's Law:

$$V = IR$$

or solving for R gives:

$$R = V/I \quad (1)$$

In this experiment, you will measure V and I to determine R for various lengths of wire. You will then make a graph of R versus length L . The resistance of a wire depends on the length of the wire, the cross-sectional area, A , and the resistivity ρ , of the material:

$$R = \rho L/A \quad (2)$$

A plot of R vs. L will result in a straight line that has a slope equal to ρ/A .

Thus the resistivity is given by

$$\rho = (\text{slope})A = (\text{slope})\pi(d/2)^2 \quad (3)$$

where d is the diameter of the wire.

1. What is resistivity, and how does it differ from resistance?

2. Why do we use different wire diameters in this experiment?

3. How does the length of a conductor affect its resistance?

4. Derive the formula for resistance and explain each term.

Answer all the questions :

Ans:

Setup

1. Make the connections as shown in Figure 1.
2. Select four brass wires of different diameter and measure* their diameters. Also choose one wire of different material e.g. Nichrome, Steel (Stainless), Aluminum, Copper and measure its diameter as well. The color code of different materials are: yellow (Brass), lighter grey (Aluminum), and dark grey (Nichrome and Stainless steel). Out of the five metals, only steel is attracted to a magnet.

*If you do not have a micrometer (or good digital calipers), use these values of diameter:

0.050 IN

0.040 IN (use this for the non-brass wires as well)

0.032 IN

0.020 IN

3. On the Resistivity apparatus, move the Reference Probe and the Slider Probe to the Park position. The probes should be as far left and right respectively as possible so the probe lifts up to allow installation of the sample wire. They will click into position.
4. Turn the two black handles counterclockwise to open the clamps to allow the sample wire to slide into position.
5. Install the copper wire in the apparatus. Slide from left or right using the white line-up hash marks. Figure 2 shows the right hand side as the wire slides in. Note that on the right hand side, the wire is on the far side of the silver clamp (with black handle), but on the left hand side the wire will be on the near side of the clamp as shown in Figure 3. This prevents the wire from bowing as you tighten the clamps.
6. Tighten the clamps by turning the black handles clockwise.

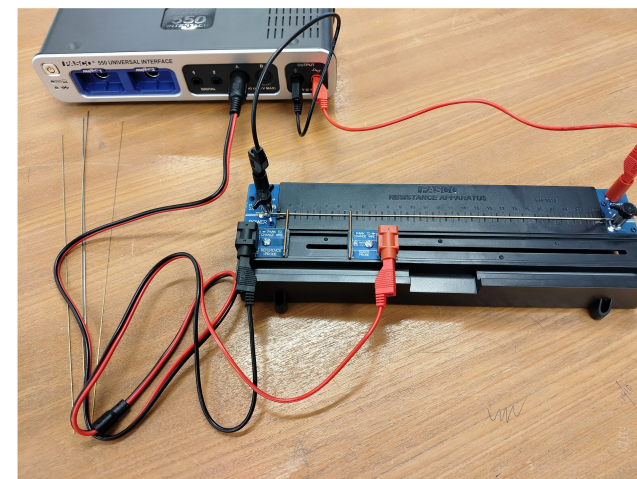


Figure 1: Apparatus Connections

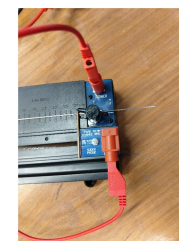


Figure 2: Right Hand Clamp

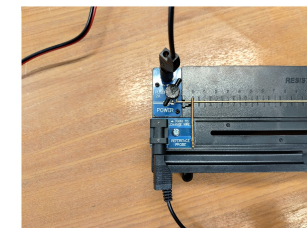


Figure 3: Left Hand Clamp

Calibration

1. Turn on the PASCO 550 Universal Interface by pressing and holding the power button for 2 sec.
2. Click open the Signal Generator (left side of screen). Set for a DC voltage of 550 Output to 0. Click the 'On' button to turn the Signal Generator on.
3. Click on 'record' (on the bottom left)
4. If the current on the 'Graph 1' is 0, no action is needed. Otherwise, go to Hardware Setup → Output Current Sensor, Ch I → Properties → Zero Sensor Now.
5. Set the DC voltage at 7.0 V. Go to the graph and set the Y-axis to 'Voltage (V)'.
6. The displayed voltage should be 0 for the Figure 4 connection. If not, reset the voltage sensor: Go to Hardware Setup → Voltage Sensor, Ch A → Properties → Zero Sensor Now
7. Stop the recording and turn off the Signal Generator

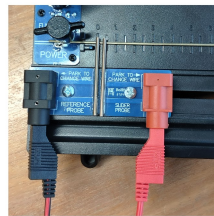
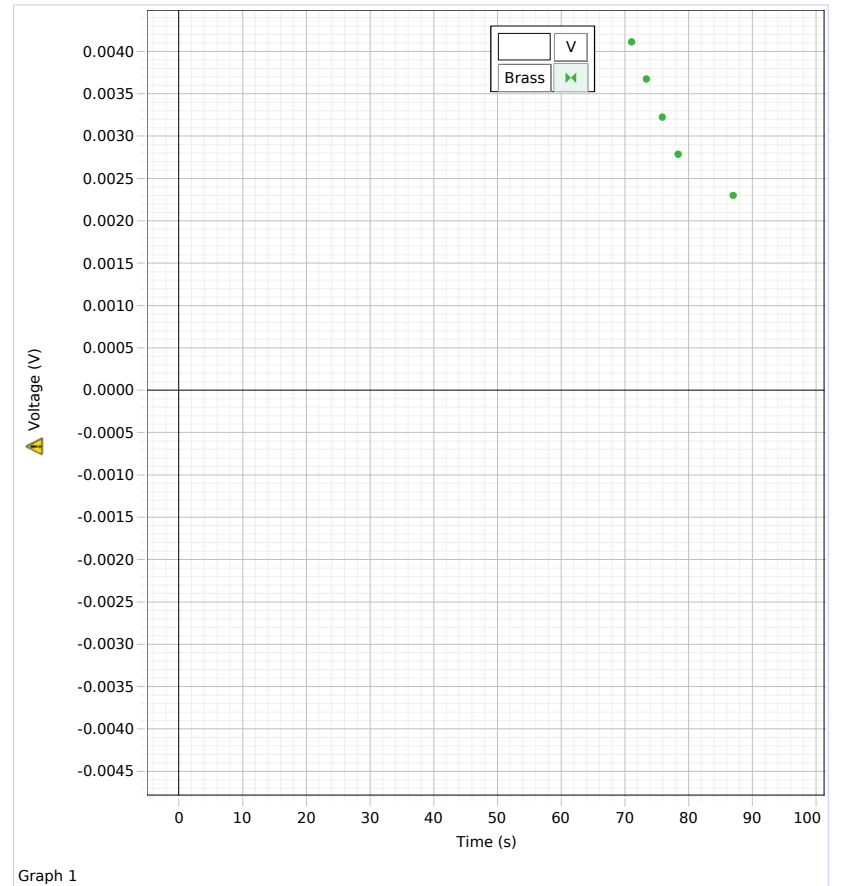


Figure 4



Graph 1

Procedure 1

1. Enable the Signal Generator:

- Go to the 'Data' tab.
- Turn on the Signal Generator and set the DC voltage to 7 V

2. Collect Data:

- Slider probe should be at 24 cm.
- Click Preview (bottom left of the screen) and wait a few seconds.
- Click 'Keep Sample', ensuring the sample corresponds to the given length.
- Change the length and repeat Keep Sample for each value.
- Turn off the signal generator after data collection

3. Analyze the Data:

- Click on 'Scale access to all data' over the graph (left most) to see all the data points.
- Once the R vs. L graph appears, set the curve fitting to 'Proportional: Ax'

4. The Data summary can show some runs. Double click on the last run and re-label it 'Nichrome'. You can delete all the other runs using the white triangle by the Delete Last Run icon at the bottom of the page.

5. Replace the Nichrome wire with the Brass wire of same diameter. Repeat steps 1 to 4 (re-label name 'Brass').

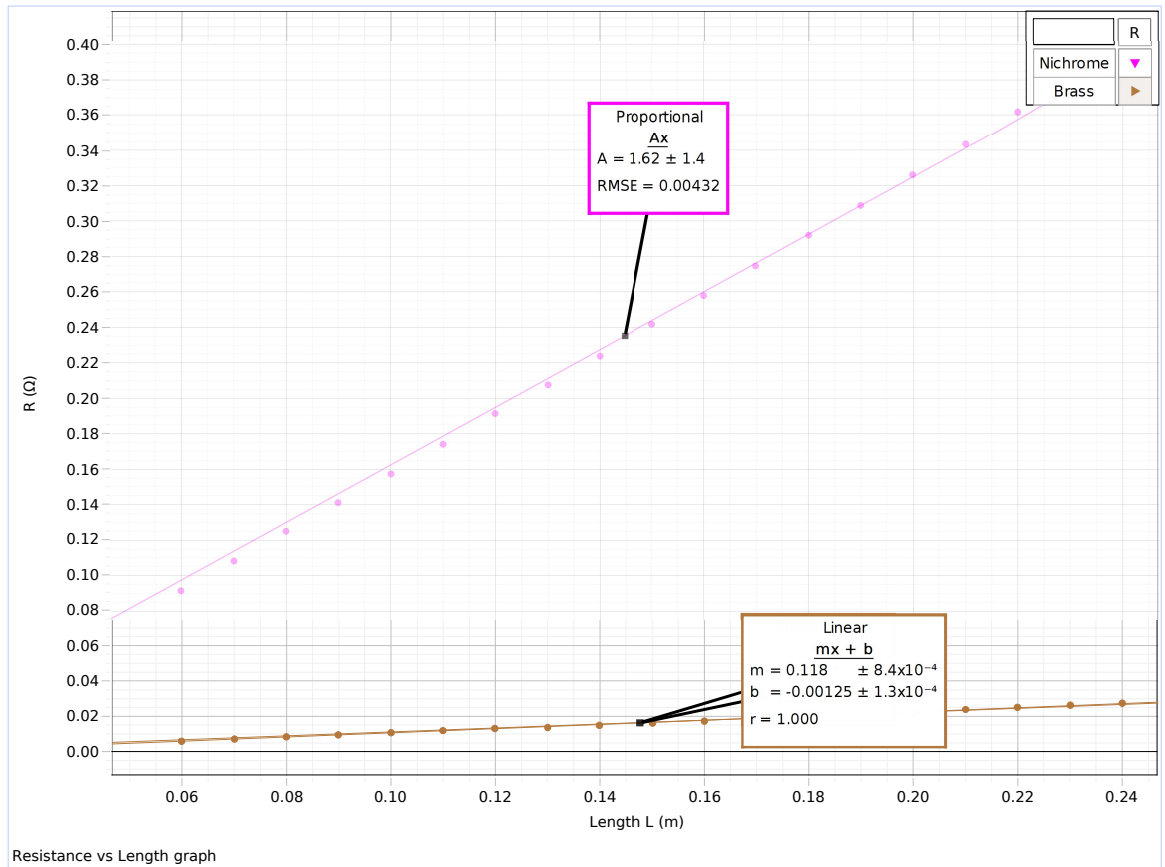
When switching between different tabs, the system will prompt you with options to either 'Turn Off' or 'Leave On.' Select **Leave On to continue.

****If you get negative value for resistance R any time, do the calibration part again.**

[Table 2: Brass wires]

	▲ Set Length L (cm) R (mΩ)	► Brass R (mΩ)
1	24	27.73
2	23	26.07
3	22	24.91
4	21	23.66
5	20	22.35
6	19	21.15
7	18	19.91
8	17	18.70
9	16	17.48
10	15	16.34
11	14	15.16
12	13	13.90
13	12	12.86
14	11	11.80
15	10	10.69
16	9	9.54
17	8	8.39
18	7	7.24
19	6	5.98
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Calculation 1 :

Form eqn 3 we can write

$$\rho = \text{slope} \pi (d/2)^2 \quad (4)$$

Diameter $d = 0.001016 \text{ m (0.04 in)}$

$$\text{Percentage Error} = |\text{calculated } \rho - \text{Standard } \rho| / \text{Standard } \rho \times 100\%$$

Standard value of ρ for Nichrome and Brass =

For Nichrome,

Slope = {Plug slope value from previous graph}
Resistivity ρ =

Percentage Error =

For Brass,

Slope =
Resistivity ρ =

Percentage Error =

Procedure 2

1. Position the Slider Probe:

- Move the Slider Probe on the Resistance Apparatus to 24.0 cm.

2. Enable the Signal Generator:

- Go to the 'Data 2' tab.
- Turn on the Signal Generator.

3. Collect Data at 24 cm (for same materials):

- Click Preview and wait a few seconds.
- Click 'Keep Sample'
- Replace the wire in the Resistance Apparatus with a brass wire of a different diameter.

4. Follow the procedure 3 for different diameters. Then turn off the signal generator

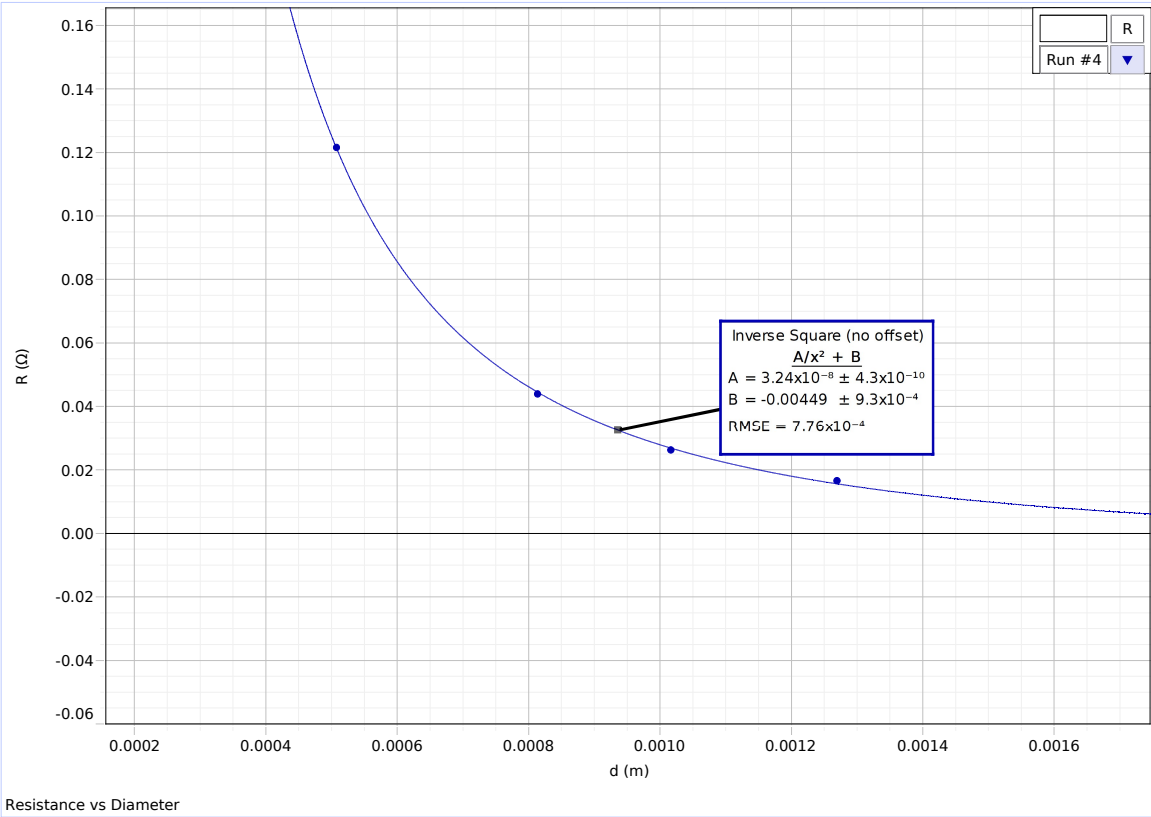
5. Analyze the Data:

- Once the R vs. d graph appears, set the curve fitting to 'Inverse Square (no offset)'

**To get the value of ρ in table for 'Data 2' tab we use the equation 3 which was inputted in calculator

Resistance Values for Various Diameters

	▲ Set	▼ Run #4	⌂ Run #4
	d (in)	R (mΩ)	ρ (Ω.m)
1	0.050	16.76	5.09560E-9
2	0.040	26.19	5.09543E-9
3	0.032	43.89	5.46579E-9
4	0.020	121.41	5.90587E-9
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Calculation

Calculate the average value from the table 2

Avg ρ for brass is =

Standard value of ρ for is =

Percentage Error =

=

1. Why is it important to zero the voltage and current sensors before taking measurements?

2. What is the purpose of the "Proportional: Ax " fitting in the R vs. L graph?

3. If two wires of different diameters but the same material are used, how will their resistance compare?

4. Why do we use different wire diameters in this experiment?

5. How can you verify whether the material follows Ohm's law using the collected data?

Answer all the questions:

Ans: